# Exercise 4 - Hotel management system

## 1. Actors

Hotel client: makes a reservation through the website or mobile app and pays online using a CIB card.

Receptionist: finds a suitable available room, checks reservations, performs check-in and checkout, closes the invoice and receives payment.

Cashier: adds the client's consumptions to the invoice.

Cleaning staff: records that a room has been cleaned, changing its status to "clean waiting".

Floor manager: checks cleaned rooms and marks them "available", or marks damaged rooms "unavailable".

Client's bank server: validates online card payments. It is an external system actor.

Maintenance staff are mentioned indirectly through repairs. The text does not say they use the application, so they are not shown as software actors. Node.js, Android, the website and the mobile app are technologies or access channels, not actors.

## 2. Actions outside the system

The phone conversation used to request a reservation; the client's physical arrival and departure; serving and consuming meals; receiving physical payment at reception; cleaning and inspecting rooms; and repairing damage all happen outside the software. Recording reservations, payments, consumptions and room-status changes happens inside it. The bank's internal payment validation takes place outside the hotel system, which exchanges payment information with the bank.

## 3. Functional boundaries

Reservations: record reservations made online or by phone.
Reception and stays: search for rooms, verify reservations, check in and check out.
Billing: process online payments, record consumptions, close invoices and record payments at checkout.
Cleaning and room status: record cleaning, approve rooms and report damage.

## 4. Use-case diagram

The diagram is on the next page. It uses ordinary actor-use-case associations. The effects of check-in and checkout are explained below instead of adding separate technical use cases.

Check-in: the selected room changes from "available" to "occupied"; a stay is created for the chosen duration, with an associated invoice.
Checkout: the stay and invoice are closed, payment is received, and the room becomes "dirty available".
After checkout: cleaning makes the room "clean waiting"; the floor manager's approval makes it "available". A damaged room stays "unavailable" until maintenance intervenes. The text does not specify who records its return to service.

# Exercise 4 - Use-case diagram

Hotel management software - web and mobile access

